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Table S1. Search Strategy by Database

database	records	search query	filters	date searched
PubMed	286	("nicotinamide mononucleotide" OR "NMN" OR "nicotinamide riboside" OR "NR"[tiab] OR "NIAGEN" OR "NAD+ precursor" OR "NAD precursor") AND ("randomized controlled trial"[pt] OR "randomized"[tiab] OR "placebo"[tiab] OR "RCT"[tiab]) AND ("insulin"[tiab] OR "glucose"[tiab] OR "HOMA"[tiab] OR "HbA1c"[tiab] OR "glycated hemoglobin"[tiab] OR "lipid"[tiab] OR "cholesterol"[tiab] OR "triglyceride"[tiab] OR "body weight"[tiab] OR "body composition"[tiab] OR "BMI"[tiab] OR "blood pressure"[tiab] OR "energy expenditure"[tiab] OR "NAD+"[tiab] OR "metabolic"[tiab] OR "fasting"[tiab] OR "C-reactive protein"[tiab] OR "interleukin"[tiab] OR "liver enzyme"[tiab] OR "ALT"[tiab] OR "AST"[tiab]); Filter: Humans, Clinical Trial or RCT	Humans; Clinical Trial or RCT publication type	2026-03-06
Embase (Ovid)	375	1: nicotinamide mononucleotide.mp. OR NMN.ti.ab. OR beta-nicotinamide mononucleotide.mp. OR MIB-626.mp. 2: nicotinamide riboside.mp. OR NIAGEN.mp. OR NR.ti.ab. 3: NAD+ precursor*.mp. OR NAD precursor*.mp. 4: 1 OR 2 OR 3; 5: insulin.ti.ab. OR glucose.ti.ab. OR HOMA*.ti.ab. OR HbA1c.ti.ab. OR glycated hemoglobin.ti.ab. OR lipid*.ti.ab. OR cholesterol.ti.ab. OR triglyceride*.ti.ab. OR body weight.ti.ab. OR body mass index.ti.ab. OR BMI.ti.ab. OR body composition.ti.ab. OR body fat.ti.ab. OR lean mass.ti.ab. OR energy expenditure.ti.ab. OR blood pressure.ti.ab. OR NAD+.ti.ab. OR C-reactive protein.ti.ab. OR CRP.ti.ab. OR interleukin*.ti.ab. OR ALT.ti.ab. OR AST.ti.ab. OR metabolic.ti.ab. 6: randomized controlled trial/ OR randomized.ti.ab. OR placebo.ti.ab. OR double-blind*.ti.ab. OR clinical trial/ OR crossover.ti.ab. 7: 4 AND 5 AND 6; 8: limit 7 to human; 9: remove duplicates from 8	Humans; remove duplicates	2026-03-06
Scopus	398	TITLE-ABS-KEY(("nicotinamide mononucleotide" OR "NMN" OR "beta-nicotinamide mononucleotide" OR "MIB-626" OR "nicotinamide riboside" OR "NIAGEN" OR "NAD+ precursor" OR "NAD precursor") AND ("randomized" OR "randomised" OR "placebo" OR "RCT" OR "clinical trial") AND ("insulin" OR "glucose" OR "HOMA" OR "HbA1c" OR "lipid" OR "cholesterol" OR "triglyceride" OR "body weight" OR "body composition" OR "BMI" OR "blood pressure" OR "energy expenditure" OR "NAD+" OR "C-reactive protein" OR "metabolic" OR "ALT" OR "AST")); Limit: Document type = Article OR Review; Source type = Journal	Article or Review; Journal source	2026-03-06
Web of Science	319	TS= (("nicotinamide mononucleotide" OR "NMN" OR "beta-nicotinamide mononucleotide" OR "MIB-626" OR "nicotinamide riboside" OR "NIAGEN" OR "NAD+ precursor" OR "NAD precursor") AND ("randomized" OR	Article document type	2026-03-06

		"randomised" OR "placebo" OR "RCT" OR "clinical trial") AND ("insulin" OR "glucose" OR "HOMA" OR "HbA1c" OR "lipid" OR "cholesterol" OR "triglyceride" OR "body weight" OR "body composition" OR "BMI" OR "blood pressure" OR "energy expenditure" OR "NAD+" OR "C-reactive protein" OR "metabolic" OR "ALT" OR "AST")); Refined by: Document Types = Article		
Cochrane CENTRAL	309	#1: "nicotinamide mononucleotide" OR "NMN" OR "beta-nicotinamide mononucleotide" OR "MIB-626"; #2: "nicotinamide riboside" OR "NR" OR "NIAGEN"; #3: "NAD+ precursor" OR "NAD precursor"; #4: #1 OR #2 OR #3; #5: "insulin" OR "glucose" OR "HOMA-IR" OR "HbA1c" OR "glycated hemoglobin" OR "lipid" OR "cholesterol" OR "triglyceride" OR "body weight" OR "body mass index" OR "BMI" OR "body composition" OR "body fat" OR "lean mass" OR "energy expenditure" OR "blood pressure" OR "NAD+" OR "C-reactive protein" OR "CRP" OR "interleukin" OR "ALT" OR "AST" OR "metabolic"; #6: #4 AND #5; #7: Limit #6 to Trials	Trials	2026-03-06
TOTAL	1687			
After deduplication	1125	Duplicates removed: 562 (DOI, PMID, title fuzzy matching)		

Note: Searches conducted on 6 March 2026 across five databases. Duplicates removed via DOI, PMID, and fuzzy title matching.

Table S2. Characteristics of Included Studies

Study	Precursor	Country	Design	Population	N_randomized	N_intervention	N_control	Dose_mg/day	Duration_weeks	Age_intervention	Age_control	Pct_female	BMI_mean	Health_status	Registration	Funding	RoB_concerns
Yoshino 2021	NMN	USA	Parallel RCT	Prediabetic obese women BMI>30	25	13	12	250	10	55	57	100	35.0	Prediabetic/insulin resistant	NCT03151239	NIH NIDDK/CTSA	Small sample; clamp insulin sensitivity primary (figures only)
Huang 2022	NMN	India	Parallel RCT	Healthy adults 40-65y, BMI 18.5-35	62	31	31	300	8.6	49.9	47.8	40.3	26.2	Healthy overweight	CTRI/2020/05/025318	EffePharm Ltd provided NMN (Uthever)	Industry funding; single-centre in India
Igarashi 2022	NMN	Japan	Parallel RCT	Healthy older men 65-71y, Japan	42	10	10	250	12	71.8	71.1	0	25.0	Healthy older men	UMIN000036321	JSPS KAKENHI/Mitsubishi Corp	Supplement prep error reduced analyzable N from 21/21 to 10/10 at 12wk; underpowered
Katayoshi 2023	NMN	Japan	Parallel RCT	Healthy adults 40-59y, BMI 18.5-29.9	34	17	17	250	12	50.4	48.1	41.2	22.5	Healthy	jRCTs031200044	Mitsubishi Corp Life Sciences & Teijin	Industry-authored study
Morifuji 2024	NMN	Japan	Parallel RCT	Healthy older adults 65-75y, Japan	59	29	30	250	12	69.3	69.2	52.5	23.1	Healthy older adults	jRCTs031200381	Meiji Holdings	Industry-authored and funded
Dollerup 2018	NR	Denmark	Parallel RCT	Obese insulin-resistant men 40-70y, Denmark	40	20	20	2000	12	45.1	42.1	0	33.3	Obese insulin-resistant	NCT02303483	Novo Nordisk Foundation/Danish Council/ChromaDex	Industry-supplied; high dose 2g/day
Conze 2019	NR	Canada	Parallel RCT (4-arm)	Healthy overweight adults 40-60y, BMI 25-30, Canada	140	32	34	1000	8	50.9	50.7	57	28.0	Healthy overweight	NCT02300740	ChromaDex (funded entire trial)	Industry-funded and authored; safety primary endpoint
Elhassan 2019	NR	UK	Crossover RCT	Healthy aged men 70-80y, UK	12	12	12	1000	3	75		0	26.6	Healthy aged men	not registered	Welcome Trust/MRC/BBSRC/ChromaDex	Small n=12; 3-week duration; experimental medicine study, not powered for clinical outcomes
Remie 2020	NR	Netherlands	Crossover	Healthy overweight/	15	13	13	1000	6	59		54	30.2	Healthy overweight/o	NCT03785899	TKI-KNAW/Stichting de Weijerhorst/NWO-	Small n=13;

			RCT	obese 45-65y sedentary, Netherlands										bese		VICI/ChromaDex	crossover; multiple outcomes without multiplicity correction
Brakedal 2022	NR	Norway	Parallel RCT	Treatment-naive PD patients, Norway	30	15	15	1000	4.3	63.9	63.7	17	NR	Parkinson's disease treatment-naive	NCT03816020	Research Council Norway/Bergen Research Foundation/ChromaDex	No metabolic outcomes; 30-day duration; PD population; patent by investigators
Ahmedi 2023	NR	USA	Crossover RCT (3-arm)	CKD moderate-severe eGFR<60, USA	25	25	25	1000	6	61		40	27.7	CKD (eGFR ~37)	NCT03579693	NIH NIDDK/Dialysis Clinics/ChromaDex	Small n=25; only 1-wk washout; CKD specific population; limited metabolic outcomes
Norheim 2024	NR	Denmark	Parallel RCT	COPD patients, Denmark	40	20	20	2000	6	72.2	71.7	62.5	26.9	COPD (ex-smokers)	NCT04990869	Novo Nordisk Foundation/Nordea/Lundbeck/Elysium Health	Focus on airway inflammation not metabolic outcomes; Elysium involvement
Orr 2024	NR	USA	Parallel RCT	MCI elderly ≥65y, 85% Hispanic, USA	22	10	10	1000	10	77.1	75.2	65	NR	MCI (mild cognitive impairment)	NCT02942888	NIH grants	Very small n=10/group; pilot; MCI not metabolic population; metabolic data only in supplements
Lin 2025	NR	USA	Parallel RCT (3-arm)	Hypertensive sedentary ≥55y, USA	54	15	16	1000	6	67.5	68.3	60.8	31.5	Hypertension	NCT04112043	NIH NIA R21	No placebo-only arm; 3 arms: NR+Ex, PL+Ex, NR alone; pilot
Bandi 2025	NR	India	Parallel RCT (4-arm)	Healthy aging adults 55-70y, BMI 24-30, India	140	34	34	500	8.6	60.1	61.0	41	26.9	Healthy aging	CTRI/2023/10/058217	Laila Nutra Pvt Ltd	Industry-funded to test their herbal product;

																	NR is active control not primary focus; 500mg dose; India
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Note: NMN, nicotinamide mononucleotide; NR, nicotinamide riboside; RCT, randomized controlled trial; BMI, body mass index; PD, Parkinson's disease; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; MCI, mild cognitive impairment; RoB, risk of bias.

Table S3. Full Data Extraction for All Included Studies and Outcomes

Study	Precursor	Outcome	N_treatment	N_control	Treatment_mean	Treatment_SD	Control_mean	Control_SD	MD	SE_MD	Unit	Design	Original_dispersion	Original_unit	Notes
Bandi 2025	NR	IL-6	34	34	4.3	2.3	5.1	2.4	-0.8	0.5701	pg/mL	Parallel RCT (4-arm)	SD	pg/mL	NS vs placebo; sig within NR group
Bandi 2025	NR	NAD+	34	34	31.9	6.5	25.8	7.1	6.1	1.6508	uM	Parallel RCT (4-arm)	SD	uM	
Conze 2019	NR	ALT	32	34	21.2	8.9	24.7	11.9	-3.5	2.5994	U/L	Parallel RCT (4-arm)	SD	U/L	
Conze 2019	NR	AST	32	34	20.8	5.5	22.2	6.6	-1.4	1.5004	U/L	Parallel RCT (4-arm)	SD	U/L	
Conze 2019	NR	HDL	32	34	54.9114	17.4015	57.2316	13.5345	-2.3202	3.8246	mg/dL	Parallel RCT (4-arm)	SD	mmol/L	
Conze 2019	NR	LDL	32	34	121.8105	31.3227	130.3179	22.8153	-8.5074	6.7164	mg/dL	Parallel RCT (4-arm)	SD	mmol/L	
Conze 2019	NR	TC	32	34	200.6973	35.5764	214.6185	29.7759	-13.9212	8.0574	mg/dL	Parallel RCT (4-arm)	SD	mmol/L	NR 1000mg vs placebo
Conze 2019	NR	TG	32	34	122.2266	61.1133	138.1692	84.1415	-15.9426	18.1986	mg/dL	Parallel RCT (4-arm)	SD	mmol/L	
Dollerup 2018	NR	ALT	20	20	29.6000	16.0997	32.0000	15.2053	-2.4000	4.9518	U/L	Parallel RCT	SEM	U/L	
Dollerup 2018	NR	BMI	20	20	32.4000	3.5777	33.3000	4.4721	-0.9000	1.2806	kg/m2	Parallel RCT	SEM	kg/m2	Baseline values; no post-treatment BMI in paper
Dollerup 2018	NR	FBG	20	20	100.9008	8.0579	104.5044	8.0579	-3.6036	2.5481	mg/dL	Parallel RCT	SEM	mmol/L	Need ×18.018 for mg/dL
Dollerup 2018	NR	HDL	20	20	46.4040	17.2937	50.2710	17.2937	-3.8670	5.4688	mg/dL	Parallel RCT	SEM	mmol/L	
Dollerup 2018	NR	HOMA-IR	20	20	2.3617	0.8944	2.5537	1.3416	-0.1920	0.3606		Parallel RCT	SEM		Derived from glucose×insulin/22.5
Dollerup 2018	NR	HbA1c	20	20	5.6000	0.4472	5.8000	0.4472	-0.2000	0.1414	%	Parallel RCT	SEM	%	From Table 3
Dollerup 2018	NR	LDL	20	20	127.6110	34.5875	127.6110	34.5875	0.0000	10.9375	mg/dL	Parallel RCT	SEM	mmol/L	
Dollerup 2018	NR	TC	20	20	204.9510	34.5875	204.9510	34.5875	0.0000	10.9375	mg/dL	Parallel RCT	SEM	mmol/L	
Dollerup 2018	NR	TG	20	20	159.4260	79.2194	132.8550	39.6097	26.5710	19.8049	mg/dL	Parallel RCT	SEM	mmol/L	TG sig increased in NR group
Dollerup 2018	NR	fasting_insulin	20	20	9.4888	3.6060	9.9064	3.6060	-0.4176	1.1403	uU/mL	Parallel RCT	SEM	pmol/L	Need /6.945 for uU/mL
Huang 2022	NMN	ALT	31	31	17.20	6.58	19.45	8.77	-2.25	1.9692	U/L	Parallel RCT	SD	U/L	CORRECTED: Paper Table 11 Day 60 values
Huang 2022	NMN	AST	31	31	18.31	4.09	18.97	4.11	-0.66	1.0414	U/L	Parallel RCT	SD	U/L	CORRECTED: Paper Table 11 Day 60 values
Huang 2022	NMN	DBP	31	31	76.81	8.30	77.84	9.01	-1.03	2.2002	mmHg	Parallel RCT	SD	mmHg	CORRECTED: Paper Table 5 Day 60 values
Huang 2022	NMN	FBG	31	31	91.53	21.41	108.48	47.12	-16.95	9.2957	mg/dL	Parallel RCT	SD	mg/dL	CORRECTED: Paper Table 7 Day

															60 values
Huang 2022	NMN	HOMA-IR	31	31	1.79	0.94	2.39	2.08	-0.6	0.41		Parallel RCT	SD		CORRECTED: Paper Table 6 Day 60 endpoint values (not change-from-baseline). Baseline HOMA-IR was balanced between groups (1.78 vs 1.83; p = 0.85); endpoint comparison is therefore valid
Huang 2022	NMN	LDL	31	31	114.45	30.75	119.42	37.76	-4.97	8.7462	mg/dL	Parallel RCT	SD	mg/dL	CORRECTED: Paper Table 13 Day 60 values
Huang 2022	NMN	NAD+	31	31	9.07	5.65	8.14	4.86	0.93	1.3385	pmol/mL	Parallel RCT	SD	pmol/mL	CORRECTED: Paper Table 2 Day 60 blood cellular NAD+/NADH. Unit was pmol/mL not ng/mL. Between-group change from baseline p=0.40 (NS). INCOMPATIBLE UNIT with Bandi (uM) - cannot be pooled for indirect comparison
Huang 2022	NMN	SBP	31	31	124.19	12.03	126.26	13.51	-2.07	3.249	mmHg	Parallel RCT	SD	mmHg	CORRECTED: Paper Table 5 Day 60 values
Huang 2022	NMN	TC	31	31	172.86	31.89	179.79	39.40	-6.93	9.1039	mg/dL	Parallel RCT	SD	mg/dL	CORRECTED: Paper Table 13 Day 60 values
Huang 2022	NMN	TG	31	31	132.48	60.41	167.91	198.89	-35.43	37.3331	mg/dL	Parallel RCT	SD	mg/dL	CORRECTED: Paper Table 13 Day 60 values
Huang 2022	NMN	fasting_insulin	31	31	14.06	7.52	19.08	18.09	-5.02	3.5186	uU/mL	Parallel RCT	SD	uU/mL	CORRECTED: Paper Table 8 Day 60 values
Katayosh i 2023	NMN	ALT	17	17	17.4000	8.3000	22.4000	9.8000	-5.0000	3.1148	U/L	Parallel RCT	SD	U/L	
Katayosh i 2023	NMN	AST	17	17	20.8000	5.4000	23.1000	6.1000	-2.3000	1.9759	U/L	Parallel RCT	SD	U/L	
Katayosh i 2023	NMN	BMI	17	17	21.9000	4.3000	21.5000	2.2000	0.4000	1.1715	kg/m2	Parallel RCT	SD	kg/m2	
Katayosh i 2023	NMN	DBP	17	17	70.8000	15.5000	79.4000	10.3000	-8.6000	4.5136	mmHg	Parallel RCT	SD	mmHg	
Katayosh i 2023	NMN	FBG	17	17	89.0000	5.6000	89.9000	6.5000	-0.9000	2.0809	mg/dL	Parallel RCT	SD	mg/dL	
Katayosh i 2023	NMN	HDL	17	17	73.5000	16.2000	74.2000	16.9000	-0.7000	5.6779	mg/dL	Parallel RCT	SD	mg/dL	
Katayosh i 2023	NMN	LDL	17	17	120.3000	24.9000	121.3000	47.6000	-1.0000	13.0289	mg/dL	Parallel RCT	SD	mg/dL	
Katayosh i 2023	NMN	SBP	17	17	119.2000	18.2000	127.1000	13.0000	-7.9000	5.4246	mmHg	Parallel RCT	SD	mmHg	
Katayosh i 2023	NMN	TG	17	17	70.6000	42.1000	83.1000	46.6000	-12.5000	15.2315	mg/dL	Parallel RCT	SD	mg/dL	

Katayoshi 2023	NMN	body_weight	17	17	61.9000	18.6000	57.6000	7.8000	4.3000	4.8918	kg	Parallel RCT	SD	kg	
Morifuji 2024	NMN	ALT	30	29	18.4000	5.6000	16.4000	5.3000	2.0000	1.4191	U/L	Parallel RCT	SD	U/L	
Morifuji 2024	NMN	AST	30	29	23.3000	5.1000	22.1000	5.0000	1.2000	1.3149	U/L	Parallel RCT	SD	U/L	
Morifuji 2024	NMN	BMI	30	29	22.2000	2.7000	22.6000	3.7000	- 0.4000	0.8456	kg/m2	Parallel RCT	SD	kg/m2	
Morifuji 2024	NMN	DBP	30	29	74.0000	11.0000	77.0000	13.0000	- 3.0000	3.1402	mmHg	Parallel RCT	SD	mmHg	
Morifuji 2024	NMN	FBG	30	29	96.3963	7.2072	93.5134	6.4865	2.8829	1.7839	mg/dL	Parallel RCT	SD	mmol/L	Need ×18.018 for mg/dL
Morifuji 2024	NMN	HDL	30	29	66.5124	14.6946	64.9656	20.1084	1.5468	4.5979	mg/dL	Parallel RCT	SD	mmol/L	
Morifuji 2024	NMN	HbA1c	30	29	5.4600	0.2700	5.4700	0.3000	- 0.0100	0.0744	%	Parallel RCT	SD	%	From Table 6
Morifuji 2024	NMN	LDL	30	29	124.1307	29.3892	123.3573	27.4557	0.7734	7.4017	mg/dL	Parallel RCT	SD	mmol/L	
Morifuji 2024	NMN	SBP	30	29	122.0000	15.0000	124.0000	19.0000	- 2.0000	4.4663	mmHg	Parallel RCT	SD	mmHg	
Morifuji 2024	NMN	TC	30	29	218.8722	38.6700	220.0323	34.0296	- 1.1601	9.4751	mg/dL	Parallel RCT	SD	mmol/L	
Morifuji 2024	NMN	TG	30	29	92.1128	48.7135	111.5982	65.5418	- 19.4854	15.0741	mg/dL	Parallel RCT	SD	mmol/L	
Morifuji 2024	NMN	body_weight	30	29	58.2000	10.6000	59.4000	11.9000	- 1.2000	2.9374	kg	Parallel RCT	SD	kg	
Morifuji 2024	NMN	fat_mass	30	29	16.6000	5.0000	17.1000	6.4000	- 0.5000	1.4986	kg	Parallel RCT	SD	kg	
Morifuji 2024	NMN	skeletal_muscle	30	29	22.3000	4.7000	22.7000	4.9000	- 0.4000	1.2507	kg	Parallel RCT	SD	kg	
Remie 2020	NR	DBP	13	13	77.9	7.2111	77.1	6.49	0.8	2.6907	mmHg	Crossover RCT	SEM	mmHg	36h ambulatory
Remie 2020	NR	FBG	13	13	98.0179	8.4454	98.7386	9.0951	- 0.7207	3.4423	mg/dL	Crossover RCT	SEM	mmol/L	Crossover end-of-period values
Remie 2020	NR	FFM_pct	13	13	62.65	8.9778	61.32	9.3023	1.33	3.5856	%	Crossover RCT	SEM	%	Significant increase
Remie 2020	NR	HDL	13	13	51.0444	12.5484	51.0444	16.7312	0.0	5.8005	mg/dL	Crossover RCT	SEM	mmol/L	
Remie 2020	NR	LDL	12	13	130.3179	22.7727	132.2514	25.0968	- 1.9335	9.613	mg/dL	Crossover RCT	SEM	mmol/L	n=12 for LDL
Remie 2020	NR	SBP	13	13	126.6	11.5378	124.6	8.6533	2.0	4.0	mmHg	Crossover RCT	SEM	mmHg	36h ambulatory
Remie 2020	NR	TC	13	13	214.6185	48.7993	214.2318	41.828	0.3867	17.826	mg/dL	Crossover RCT	SEM	mmol/L	
Remie 2020	NR	TG	13	13	144.3691	121.3506	139.0549	111.7703	5.3142	45.7574	mg/dL	Crossover RCT	SEM	mmol/L	
Remie 2020	NR	body_fat_pct	13	13	37.35	8.9778	38.68	9.3023	-1.33	3.5856	%	Crossover RCT	SEM	%	Significant decrease
Remie 2020	NR	body_weight	13	13	85.2	3.8	85.5	3.8	-0.3	1.4905	kg	Crossover RCT	SD	kg	Body weight reported as SD
Remie 2020	NR	sleeping_metabolic_rate	13	13	6.68	1.0817	6.49	1.1177	0.19	0.4314	MJ/day	Crossover RCT	SEM	MJ/day	
Yoshino 2021	NMN	BMI	13	12	33.8000	5.4083	33.3000	3.1177	0.5000	1.7493	kg/m2	Parallel RCT	SEM	kg/m2	
Yoshino 2021	NMN	DBP	13	12	75.0000	10.8167	77.0000	10.3923	- 2.0000	4.2426	mmHg	Parallel RCT	SEM	mmHg	
Yoshino 2021	NMN	FBG	13	12	100.9008	12.9930	100.9008	12.4832	0.0000	5.0963	mg/dL	Parallel RCT	SEM	mmol/L	Need ×18.018 for mg/dL
Yoshino 2021	NMN	HDL	13	12	50.2710	13.9427	51.8178	9.3770	- 1.5468	4.7203	mg/dL	Parallel RCT	SEM	mmol/L	

Yoshino 2021	NMN	HbA1c	13	12	5.5000	0.3606	5.5000	0.3464	0.0000	0.1414	%	Parallel RCT	SEM	%	From Table 1 After
Yoshino 2021	NMN	SBP	13	12	125.0000	18.0278	131.0000	13.8564	- 6.0000	6.4031	mmHg	Parallel RCT	SEM	mmHg	
Yoshino 2021	NMN	TG	13	12	111.5982	57.4819	131.0836	70.5676	- 19.4854	25.8679	mg/dL	Parallel RCT	SEM	mmol/L	
Yoshino 2021	NMN	body_weight	13	12	89.0000	14.4222	87.0000	10.3923	2.0000	5.0000	kg	Parallel RCT	SEM	kg	
Yoshino 2021	NMN	fasting_insulin	13	12	15.8000	9.7350	17.2000	8.6603	- 1.4000	3.6797	uU/mL	Parallel RCT	SEM	uU/mL	

Note: MD, mean difference; SE, standard error; SD, standard deviation; SEM, standard error of the mean. Values represent post-intervention group means. Crossover SDs estimated from reported statistics where necessary.

Table S4. Risk of Bias 2 (RoB 2) Domain-Level Assessments

Study	D1_Randomization	D1_Justification	D2_Deviations	D2_Justification	D3_Missing_data	D3_Justification	D4_Measurement	D4_Justification	D5_Reporting	D5_Justification	Overall
Yoshino 2021	Some concerns	No explicit detail on sequence generation or allocation concealment in text	Low	DB-PC with matching placebo; all 25 completed; analysis consistent with assignment	Low	25/25 analyzed; no missing data	Low	DB; all metabolic outcomes objective lab measures	Some concerns	Pre-registered NCT03151239, but metabolic outcomes are secondary; Yoshino holds NMN patent; Imai SAB Metro Biotech	Some concerns
Huang 2022	Some concerns	Randomized 1:1, but post-unblinding exclusion of 4 placebo subjects to equalize groups is non-standard	Low	DB-PC with matching capsules; no protocol deviations	Some concerns	66 randomized - > 62 analyzed; post-unblinding exclusion of 4 subjects is methodologically problematic	Low	DB; all outcomes objective lab measures	Some concerns	Registered CTRI; HOMA-IR exploratory; industry-funded by EffePharm (NMN manufacturer)	Some concerns
Igarashi 2022	Low	Independent third-party allocation (C&C Inc.); sealed randomization code; no baseline differences	Some concerns	DB-PC but supplement preparation error meant some participants received incorrect supplements	High	42 randomized but only 10/10 analyzed at 12wk - 52% attrition from supplement error	Some concerns	DB; objective lab outcomes; but most metabolic data only in supplementary tables	Some concerns	Registered UMIN; metabolic outcomes secondary/exploratory; industry-supplied NMN (Mitsubishi); data relegated to supplements	High
Katayoshi 2023	Low	Block random allocation by independent controller (Orthomedico Inc.); allocation concealed until unblinding	Low	DB; indistinguishable capsules; 34/36 analyzed; consistent with assignment	Low	36 enrolled, 34 analyzed (5.6% attrition); 2 excluded per pre-specified criteria	Low	DB; objective blood biomarkers analyzed by outsourced lab (BML)	Some concerns	Registered jRCTs; all authors are Mitsubishi/Teijin employees (manufacturers); NAD+ below LOQ buried	Some concerns
Morifuji 2024	Some concerns	Described as randomized DB-PC but specific sequence generation and allocation concealment not detailed	Low	DB-PC with matching placebo; 59/59 analyzed; no deviations	Low	59/59 analyzed; zero attrition; complete data	Low	DB; all outcomes objective lab measures incl DXA	Some concerns	Registered jRCTs; all authors Meiji Holdings employees (NMN manufacturer and funder)	Some concerns
Dollerup 2018	Low	Aarhus University Hospital Pharmacy managed randomization/blinding/packaging - gold standard	Low	DB; identical capsules; all 40 completed; no deviations	Low	40/40 completed; zero attrition	Low	DB; objective outcomes (clamp/OGTT/DXA/labs); same procedures throughout	Some concerns	Pre-registered NCT02303483; 2019b is secondary publication; ChromaDex supplied; Brenner adviser/stockholder	Some concerns
Conze 2019	Some concerns	Randomized 4-arm registered but specific sequence generation/allocation concealment not detailed	Low	DB; identical NIAGEN and placebo capsules; 133/140 completed (95%)	Low	140 randomized 133 completed; 7 dropouts balanced across arms	Low	DB; objective safety/metabolic lab measures	Some concerns	Pre-registered NCT; ChromaDex funded entire trial; 2 authors employees; Brenner adviser/stockholder; metabolic outcomes secondary	Some concerns
Elhassan 2019	Some concerns	Crossover RCT DB but NOT registered; no details on sequence generation or allocation concealment	Some concerns	DB crossover; all 12 completed; but carry-over effects noted for some cytokines suggesting insufficient 21d washout	Low	12/12 completed both phases; no missing data	Some concerns	DB; objective labs; but metabolic data only in Supp Table S1 with main text stating "no changes" narratively	High	NOT pre-registered; no protocol to verify pre-specified outcomes; metabolic data relegated to supplements; Brenner has financial interests	High
Remie 2020	Low	Computer-generated random numbers; clearly described; registered NCT03785899	Low	DB crossover; 4-7wk washout adequate; no	Some concerns	15 randomized, 13 completed (87%); 2	Low	DB; objective outcomes (lab/DXA/clamp); same procedures each phase	Some concerns	Pre-registered; authors acknowledge multiple testing without	Some concerns

				deviations		dropouts in small crossover could affect precision				correction and "false positives possible"	
Brakedal 2022	Low	Block randomization (block size 6) via eCRF; independent third party managed allocation/packing	Low	DB; identical capsules; all 30 completed the 30d intervention	Low	30/30 completed; zero attrition	Low	DB; objective neuroimaging and clinical scales by blinded personnel	Some concerns	Pre-registered NCT; but Tzoulis/Dolle filed patent on NR in PD; post-hoc responder analyses	Some concerns
Ahmadi 2023	Some concerns	3-arm crossover registered NCT03579693 but specific randomization method not described	Some concerns	DB crossover; all 25 completed; but 1-week washout is pharmacokinetically short for 6-week treatment periods	Low	25/25 completed all 3 phases; no missing data	Some concerns	DB; objective outcomes; but most results as P-values only without means±SD	Some concerns	Pre-registered; primary outcome negative; limited metabolic data across two publications (2023/2025)	Some concerns
Norheim 2024	Low	Permuted block randomization (block 4) stratified by CAT score; adequate	Low	DB; matching capsules; 37/40 analyzed; no deviations	Low	40 randomized, 37 analyzed (92.5%); balanced dropout reasons	Low	DB; objective outcomes; LS means from mixed models - valid method	Some concerns	Pre-registered NCT; Elysium Health provided capsules/funded; 2 employees co-authored	Some concerns
Orr 2024	Some concerns	Registered NCT02942888 but no randomization details; very small n=10/group	Low	DB with dose escalation (250->1000mg) pre-specified and uniform	Some concerns	22 randomized, 20 analyzed; 2 dropouts (9%) in very small pilot	Some concerns	DB; objective labs; but metabolic data only in supplements dismissed as "no meaningful differences"	Some concerns	Pre-registered; Brenner co-author, inventor, adviser, stockholder ChromaDex - substantial personal COI	Some concerns
Lin 2025	Some concerns	3-arm parallel registered NCT04112043 but sequence generation/concealment not described; no placebo-only arm	Low	DB; 90% supplement adherence; 93% exercise adherence; no deviations	Low	54 randomized, 49 completed (91%); balanced attrition	Low	DB; 24h ambulatory BP monitoring - validated objective measure	Some concerns	Pre-registered; NIH-funded; but pilot with no placebo-only arm complicates NMA extraction	Some concerns
Bandi 2025	Low	SAS PROC PLAN block randomization; independent statistician; adequate concealment	Low	DB; identical capsules across 4 arms; blinded until database lock	Some concerns	140 randomized, 137 PP analyzed; per-protocol only with no ITT reported	Low	DB; objective outcomes (NAD+ HPLC; cytokines ELISA; clinical measures)	Some concerns	Registered CTRI; funded by Laila Nutra (LN22199 manufacturer); NR is comparator not primary interest	Some concerns

Note: D1, bias from the randomization process; D2, bias due to deviations from intended interventions; D3, bias due to missing outcome data; D4, bias in measurement of the outcome; D5, bias in selection of the reported result. Assessments performed independently by two reviewers with disagreements resolved by discussion.

Table S5. Pairwise Meta-Analysis Results for All Outcomes

outcome	precursor	comparison	k	MD random	lower CI	upper CI	p random	I2 pct	tau2	MD fixed	lower fixed	upper fixed	p fixed	note
ALT	NMN	NMN vs Placebo	3	-1.133	-5.1349	2.869	0.579	65.8	8.0335	-0.1195	-2.2361	1.9971	0.9119	
ALT	NR	NR vs Placebo	2	-3.2624	-7.7734	1.2487	0.1563	0.0	0.0	-3.2624	-7.7734	1.2487	0.1563	
AST	NMN	NMN vs Placebo	3	-0.303	-1.9856	1.3795	0.7241	18.2	0.426	-0.2867	-1.7655	1.1921	0.704	
AST	NR	NR vs Placebo	1	-1.4	-4.3408	1.5408	0.3508	0.0	0.0	-1.4	-4.3408	1.5408	0.3508	[k=1: single study, no meta-analysis]
BMI	NMN	NMN vs Placebo	3	-0.0426	-1.2938	1.2086	0.9468	0.0	0.0	-0.0426	-1.2938	1.2086	0.9468	
BMI	NR	NR vs Placebo	1	-0.9	-3.41	1.61	0.4822	0.0	0.0	-0.9	-3.41	1.61	0.4822	[k=1: single study, no meta-analysis]
DBP	NMN	NMN vs Placebo	4	-2.5453	-5.5965	0.506	0.1021	0.0	0.0	-2.5453	-5.5965	0.506	0.1021	
DBP	NR	NR vs Placebo	1	0.8	-4.4738	6.0738	0.7662	0.0	0.0	0.8	-4.4738	6.0738	0.7662	[k=1: single study, no meta-analysis]
FBG	NMN	NMN vs Placebo	4	0.0243	-4.249	4.2975	0.9911	47.4	8.0118	0.8432	-1.6972	3.3836	0.5153	
FBG	NR	NR vs Placebo	2	-2.5831	-6.5973	1.4311	0.2072	0.0	0.0	-2.5831	-6.5973	1.4311	0.2072	
FFM_pct	NR	NR vs Placebo	1	1.33	-5.6978	8.3578	0.7107	0.0	0.0	1.33	-5.6978	8.3578	0.7107	[k=1: single study, no meta-analysis]
HDL	NMN	NMN vs Placebo	3	-0.1458	-5.7299	5.4382	0.9592	0.0	0.0	-0.1458	-5.7299	5.4382	0.9592	
HDL	NR	NR vs Placebo	3	-2.1891	-7.5936	3.2154	0.4273	0.0	0.0	-2.1891	-7.5936	3.2154	0.4273	
HOMA-IR	NMN	NMN vs Placebo	1	-0.6	-1.4036	0.2036	0.1434	0.0	0.0	-0.6	-1.4036	0.2036	0.1434	[k=1: single study, no meta-analysis]
HOMA-IR	NR	NR vs Placebo	1	-0.192	-0.8988	0.5148	0.5944	0.0	0.0	-0.192	-0.8988	0.5148	0.5944	[k=1: single study, no meta-analysis]
HbA1c	NMN	NMN vs Placebo	2	-0.0078	-0.1369	0.1212	0.9053	0.0	0.0	-0.0078	-0.1369	0.1212	0.9053	
HbA1c	NR	NR vs Placebo	1	-0.2	-0.4771	0.0771	0.1572	0.0	0.0	-0.2	-0.4771	0.0771	0.1572	[k=1: single study, no meta-analysis]
IL-6	NR	NR vs Placebo	1	-0.8	-1.9174	0.3174	0.1605	0.0	0.0	-0.8	-1.9174	0.3174	0.1605	[k=1: single study, no meta-analysis]
LDL	NMN	NMN vs Placebo	3	-1.5247	-11.6846	8.6351	0.7686	0.0	0.0	-1.5247	-11.6846	8.6351	0.7686	
LDL	NR	NR vs Placebo	3	-5.0671	-14.7059	4.5718	0.3028	0.0	0.0	-5.0671	-14.7059	4.5718	0.3028	
NAD+	NMN	NMN vs Placebo	1	0.93	-1.6935	3.5535	0.4872	0.0	0.0	0.93	-1.6935	3.5535	0.4872	[k=1: single study, no

														meta-analysis]
NAD+	NR	NR vs Placebo	1	6.1	2.8644	9.3356	0.0002	0.0	0.0	6.1	2.8644	9.3356	0.0002	[k=1: single study, no meta-analysis]
SBP	NMN	NMN vs Placebo	4	-3.4992	-7.8469	0.8484	0.1147	0.0	0.0	-3.4992	-7.8469	0.8484	0.1147	
SBP	NR	NR vs Placebo	1	2.0	-5.84	9.84	0.6171	0.0	0.0	2.0	-5.84	9.84	0.6171	[k=1: single study, no meta-analysis]
TC	NMN	NMN vs Placebo	2	-4.1603	-17.0272	8.7066	0.5263	0.0	0.0	-4.1603	-17.0272	8.7066	0.5263	
TC	NR	NR vs Placebo	3	-7.9234	-19.8717	4.0248	0.1937	0.0	0.0	-7.9234	-19.8717	4.0248	0.1937	
TG	NMN	NMN vs Placebo	4	-17.7762	-36.5297	0.9773	0.0632	0.0	0.0	-17.7762	-36.5297	0.9773	0.0632	
TG	NR	NR vs Placebo	3	4.1657	-25.3794	33.7108	0.7823	20.0	145.8899	3.6622	-21.5438	28.8681	0.7758	
body_fat_pct	NR	NR vs Placebo	1	-1.33	-8.3578	5.6978	0.7107	0.0	0.0	-1.33	-8.3578	5.6978	0.7107	[k=1: single study, no meta-analysis]
body_weight	NMN	NMN vs Placebo	3	0.6101	-3.7981	5.0184	0.7862	0.0	0.0	0.6101	-3.7981	5.0184	0.7862	
body_weight	NR	NR vs Placebo	1	-0.3	-3.2214	2.6214	0.8405	0.0	0.0	-0.3	-3.2214	2.6214	0.8405	[k=1: single study, no meta-analysis]
fasting_insulin	NMN	NMN vs Placebo	2	-3.291	-8.2754	1.6934	0.1956	0.0	0.0	-3.291	-8.2754	1.6934	0.1956	
fasting_insulin	NR	NR vs Placebo	1	-0.4176	-2.6526	1.8174	0.7142	0.0	0.0	-0.4176	-2.6526	1.8174	0.7142	[k=1: single study, no meta-analysis]
fat_mass	NMN	NMN vs Placebo	1	-0.5	-3.4373	2.4373	0.7386	0.0	0.0	-0.5	-3.4373	2.4373	0.7386	[k=1: single study, no meta-analysis]
skeletal_muscle	NMN	NMN vs Placebo	1	-0.4	-2.8514	2.0514	0.7491	0.0	0.0	-0.4	-2.8514	2.0514	0.7491	[k=1: single study, no meta-analysis]
sleeping_metabolic_rate	NR	NR vs Placebo	1	0.19	-0.6555	1.0355	0.6596	0.0	0.0	0.19	-0.6555	1.0355	0.6596	[k=1: single study, no meta-analysis]

Note: MD, mean difference; CI, confidence interval; k, number of studies. Random-effects model (DerSimonian-Laird); fixed-effect model (inverse-variance). Heterogeneity quantified by I-squared and tau-squared.

Table S6. Network Meta-Analysis Results (Direct, Indirect, and Network Estimates)

outcome	comparison	MD	lower CI	upper CI	p value	k	type	note
ALT	NMN vs Placebo	-1.133	-5.1349	2.869	0.579	3	direct	
ALT	NR vs Placebo	-3.2624	-7.7734	1.2487	0.1563	2	direct	
ALT	NMN vs NR	2.1294	-3.901	8.1598	0.4889	3+2	indirect	
AST	NMN vs Placebo	-0.303	-1.9856	1.3795	0.7241	3	direct	
AST	NR vs Placebo	-1.4	-4.3408	1.5408	0.3508	1	direct	Single study - no meta-analysis
AST	NMN vs NR	1.097	-2.2911	4.485	0.5257	3+1	indirect	Caution: NR arm k=1
BMI	NMN vs Placebo	-0.0426	-1.2938	1.2086	0.9468	3	direct	
BMI	NR vs Placebo	-0.9	-3.41	1.61	0.4822	1	direct	Single study - no meta-analysis
BMI	NMN vs NR	0.8574	-1.9471	3.6619	0.549	3+1	indirect	Caution: NR arm k=1
DBP	NMN vs Placebo	-2.5453	-5.5965	0.506	0.1021	4	direct	
DBP	NR vs Placebo	0.8	-4.4738	6.0738	0.7662	1	direct	Single study - no meta-analysis
DBP	NMN vs NR	-3.3453	-9.4381	2.7476	0.2819	4+1	indirect	Caution: NR arm k=1
FBG	NMN vs Placebo	0.0243	-4.249	4.2975	0.9911	4	direct	
FBG	NR vs Placebo	-2.5831	-6.5973	1.4311	0.2072	2	direct	
FBG	NMN vs NR	2.6074	-3.2556	8.4703	0.3834	4+2	indirect	
HDL	NMN vs Placebo	-0.1458	-5.7299	5.4382	0.9592	3	direct	
HDL	NR vs Placebo	-2.1891	-7.5936	3.2154	0.4273	3	direct	
HDL	NMN vs NR	2.0433	-5.7278	9.8144	0.6063	3+3	indirect	
HOMA-IR	NMN vs Placebo	-0.6	-1.4036	0.2036	0.1434	1	direct	Single study - no meta-analysis
HOMA-IR	NR vs Placebo	-0.192	-0.8988	0.5148	0.5944	1	direct	Single study - no meta-analysis
HOMA-IR	NMN vs NR	-0.408	-1.4782	0.6622	0.4549	1+1	indirect	Caution: NMN arm k=1, NR arm k=1
HbA1c	NMN vs Placebo	-0.0078	-0.1369	0.1212	0.9053	2	direct	
HbA1c	NR vs Placebo	-0.2	-0.4771	0.0771	0.1572	1	direct	Single study - no meta-analysis
HbA1c	NMN vs NR	0.1922	-0.1135	0.4979	0.2179	2+1	indirect	Caution: NR arm k=1
LDL	NMN vs Placebo	-1.5247	-11.6846	8.6351	0.7686	3	direct	
LDL	NR vs Placebo	-5.0671	-14.7059	4.5718	0.3028	3	direct	
LDL	NMN vs NR	3.5423	-10.4623	17.547	0.6201	3+3	indirect	
SBP	NMN vs Placebo	-3.4992	-7.8469	0.8484	0.1147	4	direct	
SBP	NR vs Placebo	2.0	-5.84	9.84	0.6171	1	direct	Single study - no meta-analysis
SBP	NMN vs NR	-5.4992	-14.464	3.4656	0.2292	4+1	indirect	Caution: NR arm k=1
TC	NMN vs Placebo	-4.1603	-17.0272	8.7066	0.5263	2	direct	
TC	NR vs Placebo	-7.9234	-19.8717	4.0248	0.1937	3	direct	
TC	NMN vs NR	3.7631	-13.7958	21.3221	0.6744	2+3	indirect	
TG	NMN vs Placebo	-17.7762	-36.5297	0.9773	0.0632	4	direct	
TG	NR vs Placebo	4.1657	-25.3794	33.7108	0.7823	3	direct	
TG	NMN vs NR	-21.9419	-56.9363	13.0525	0.2191	4+3	indirect	
body_weight	NMN vs Placebo	0.6101	-3.7981	5.0184	0.7862	3	direct	
body_weight	NR vs Placebo	-0.3	-3.2214	2.6214	0.8405	1	direct	Single study - no meta-analysis

body_weight	NMN vs NR	0.9101	-4.3783	6.1985	0.7359	3+1	indirect	Caution: NR arm k=1
fasting_insulin	NMN vs Placebo	-3.291	-8.2754	1.6934	0.1956	2	direct	
fasting_insulin	NR vs Placebo	-0.4176	-2.6526	1.8174	0.7142	1	direct	Single study - no meta-analysis
fasting_insulin	NMN vs NR	-2.8734	-8.3359	2.5892	0.3025	2+1	indirect	Caution: NR arm k=1

Note: MD, mean difference; CI, confidence interval; k, number of contributing studies. Indirect comparisons estimated via the Bucher method. Network estimates combine direct and indirect evidence where applicable.

Table S7. Leave-One-Out Sensitivity Analysis Summary

type	outcome	comparison	n_loo	full MD	full p	loo MD range	direction changes	significance changes	robust	note
Pairwise	ALT	NMN vs Placebo	3	-1.1330	0.5790	-3.0353 to 0.0943	1	0	No	
Pairwise	ALT	NR vs Placebo	2	-3.2624	0.1563	-3.5000 to -2.4000	0	0	Yes	
Pairwise	AST	NMN vs Placebo	3	-0.3030	0.7241	-1.0165 to 0.0968	1	0	No	
Pairwise	BMI	NMN vs Placebo	3	-0.0426	0.9468	-0.2295 to 0.4310	1	0	No	
Pairwise	DBP	NMN vs Placebo	4	-2.5453	0.1021	-4.0644 to -1.7278	0	0	Yes	
Pairwise	FBG	NMN vs Placebo	4	0.0243	0.9911	-2.2632 to 1.1960	3	0	No	
Pairwise	FBG	NR vs Placebo	2	-2.5831	0.2072	-3.6036 to -0.7207	0	0	Yes	
Pairwise	HDL	NMN vs Placebo	3	-0.1458	0.9592	-1.2007 to 0.6570	2	0	No	
Pairwise	HDL	NR vs Placebo	3	-2.1891	0.4273	-2.8282 to -1.6171	0	0	Yes	
Pairwise	HbA1c	NMN vs Placebo	2	-0.0078	0.9053	-0.0100 to 0.0000	0	0	Yes	
Pairwise	LDL	NMN vs Placebo	3	-1.5247	0.7686	-3.7367 to 0.3407	1	0	No	
Pairwise	LDL	NR vs Placebo	3	-5.0671	0.3028	-6.3510 to -1.0909	0	0	Yes	
Pairwise	SBP	NMN vs Placebo	4	-3.4992	0.1147	-4.7470 to -2.6156	0	0	Yes	
Pairwise	TC	NMN vs Placebo	2	-4.1603	0.5263	-6.9300 to -1.1601	0	0	Yes	
Pairwise	TC	NR vs Placebo	3	-7.9234	0.1937	-11.4939 to 0.1058	1	0	No	
Pairwise	TG	NMN vs Placebo	4	-17.7762	0.0632	-21.2154 to -16.5351	0	0	Yes	
Pairwise	TG	NR vs Placebo	3	4.1657	0.7823	-13.0394 to 23.2171	1	0	No	
Pairwise	body_weight	NMN vs Placebo	3	0.6101	0.7862	-0.3789 to 3.1752	1	0	No	
Pairwise	fasting_insulin	NMN vs Placebo	2	-3.2910	0.1956	-5.0200 to -1.4000	0	0	Yes	
Indirect (NMN vs NR)	ALT	NMN vs NR	5	2.1294	0.4889	0.2271 to 3.3566		0	Yes	
Indirect (NMN vs NR)	AST	NMN vs NR	3	1.0970	0.5257	0.3835 to 1.4968		0	Yes	Only NMN arm (k=3) varied; other arm k=1
Indirect (NMN vs NR)	BMI	NMN vs NR	3	0.8574	0.5490	0.6705 to 1.3310		0	Yes	Only NMN arm (k=3) varied; other arm k=1
Indirect (NMN vs NR)	DBP	NMN vs NR	4	-3.3453	0.2819	-4.8644 to -2.5278		0	Yes	Only NMN arm (k=4) varied; other arm k=1
Indirect (NMN vs NR)	FBG	NMN vs NR	6	2.6074	0.3834	0.3199 to 3.7791		0	Yes	
Indirect (NMN vs NR)	HDL	NMN vs NR	6	2.0433	0.6063	0.9884 to 2.8461		0	Yes	
Indirect (NMN vs NR)	HOMA-IR	NMN vs NR	0	-0.4080	0.4549	-			-	k=1 each arm; LOO not possible
Indirect (NMN vs NR)	HbA1c	NMN vs NR	2	0.1922	0.2179	0.1900 to 0.2000		0	Yes	Only NMN arm (k=2) varied; other arm k=1
Indirect (NMN vs NR)	LDL	NMN vs NR	6	3.5423	0.6201	-0.4339 to 5.4078		0	Yes	
Indirect (NMN vs NR)	SBP	NMN vs NR	4	-5.4992	0.2292	-6.7470 to -4.6156		0	Yes	Only NMN arm (k=4) varied; other arm k=1
Indirect (NMN vs NR)	TC	NMN vs NR	5	3.7631	0.6744	-4.2660 to 7.3336		0	Yes	
Indirect (NMN vs NR)	TG	NMN vs NR	7	-21.9419	0.2191	-40.9933 to -4.7368		1	No	
Indirect (NMN vs NR)	body_weight	NMN vs NR	3	0.9101	0.7359	-0.0789 to 3.4752		0	Yes	Only NMN arm

NR)										(k=3) varied; other arm k=1
Indirect (NMN vs NR)	fasting_insulin	NMN vs NR	2	-2.8734	0.3025	-4.6024 to -0.9824		0	Yes	Only NMN arm (k=2) varied; other arm k=1

Note: LOO, leave-one-out; MD, mean difference. Direction change indicates the pooled MD changed sign upon study omission. Significance change indicates the pooled p-value crossed 0.05. Analyses with $k < 3$ not performed.

Table S8. GRADE/CINeMA Certainty of Evidence Assessment for Indirect Comparisons (NMN vs NR)

Outcome	k NMN	k NR	MD 95CI	p value	Within study bias	Reporting bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Overall certainty	Downgrades
ALT	3	2	2.13 [-3.90, 8.16]	0.4889	Some concerns	Some concerns	Some concerns	Some concerns	Some concerns	Not applicable	Very low	5
AST	3	1	1.10 [-2.29, 4.49]	0.5257	Some concerns	Some concerns	Some concerns	Some concerns	No concerns	Not applicable	Very low	4
BMI	3	1	0.86 [-1.95, 3.66]	0.5490	Some concerns	Some concerns	Some concerns	Some concerns	No concerns	Not applicable	Very low	4
Diastolic BP	4	1	-3.35 [-9.44, 2.75]	0.2819	Some concerns	Some concerns	Some concerns	Some concerns	No concerns	Not applicable	Very low	4
Fasting Blood Glucose	4	2	2.61 [-3.26, 8.47]	0.3834	Some concerns	Some concerns	Some concerns	Some concerns	Low concern	Not applicable	Very low	4
HDL Cholesterol	3	3	2.04 [-5.73, 9.81]	0.6063	Some concerns	Some concerns	Some concerns	Some concerns	No concerns	Not applicable	Very low	4
HOMA-IR	1	1	-0.41 [-1.48, 0.66]	0.4549	Some concerns	Some concerns	Some concerns	Some concerns	Some concerns	Not applicable	Very low	5
HbA1c	2	1	0.19 [-0.11, 0.50]	0.2179	Some concerns	Some concerns	Some concerns	Some concerns	No concerns	Not applicable	Very low	4
LDL Cholesterol	3	3	3.54 [-10.46, 17.55]	0.6201	Some concerns	Some concerns	Some concerns	Major concerns	No concerns	Not applicable	Very low	5
Systolic BP	4	1	-5.50 [-14.46, 3.47]	0.2292	Some concerns	Some concerns	Some concerns	Some concerns	No concerns	Not applicable	Very low	4
Total Cholesterol	2	3	3.76 [-13.80, 21.32]	0.6744	Some concerns	Some concerns	Some concerns	Major concerns	No concerns	Not applicable	Very low	5
Triglycerides	4	3	-21.94 [-56.94, 13.05]	0.2191	Some concerns	Some concerns	Some concerns	Major concerns	No concerns	Not applicable	Very low	5
Body Weight	3	1	0.91 [-4.38, 6.20]	0.7359	Some concerns	Some concerns	Some concerns	Some concerns	No concerns	Not applicable	Very low	4
Fasting Insulin	2	1	-2.87 [-8.34, 2.59]	0.3025	Some concerns	Some concerns	Some concerns	Some concerns	No concerns	Not applicable	Very low	4

Note: GRADE, Grading of Recommendations Assessment, Development and Evaluation; CINeMA, Confidence in Network Meta-Analysis. All indirect comparisons started at low certainty (observational-equivalent for indirect evidence) and were further downgraded based on within-study bias, reporting bias, indirectness, imprecision, heterogeneity, and incoherence.

Table S9. Inter-Rater Agreement Across Review Phases

Phase	Items assessed	Agreements	Percent agreement	Cohen kappa	Disagreement resolution
title/abstract screening	1003	971	96.8	0.92	discussion or FAD tiebreak
fulltext screening	186	169	90.9	0.84	discussion or FAD tiebreak
RoB2 (domain-level)	75	65	86.7	0.75	discussion
data extraction	15	14	93.3	n/a	checked against source

Note: Cohen's kappa interpreted as: >0.80 almost perfect, 0.61-0.80 substantial, 0.41-0.60 moderate. FAD, final arbiter decision. All disagreements resolved by discussion between reviewers or third-party adjudication.

Table S10. Publication Bias Assessment (Funnel Plot Asymmetry and Egger's Test)

outcome	precursor	k	funnel_plot	egger_intercept	egger_se	egger_t	egger_df	egger_p	interpretation
ALT	NMN	3	funnel_ALT_NMN_vs_PBO.png	-4.4866	1.6170	-2.7746	1	0.2202	Egger's test non-significant (p = 0.220); k = 3 provides insufficient power to detect asymmetry
ALT	NR	2	Not generated (k < 3)						k = 2: insufficient for publication bias assessment
AST	NMN	3	funnel_AST_NMN_vs_PBO.png	-1.3941	3.4457	-0.4046	1	0.7552	Egger's test non-significant (p = 0.755); k = 3 provides insufficient power to detect asymmetry
AST	NR	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
BMI	NMN	3	funnel_BMI_NMN_vs_PBO.png	1.1608	0.6924	1.6765	1	0.3424	Egger's test non-significant (p = 0.342); k = 3 provides insufficient power to detect asymmetry
BMI	NR	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
DBP	NMN	4	funnel_DBP_NMN_vs_PBO.png	-2.0137	1.2045	-1.6718	2	0.2365	Egger's test non-significant (p = 0.237); k = 4 provides insufficient power to detect asymmetry
DBP	NR	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
FBG	NMN	4	funnel_FBG_NMN_vs_PBO.png	-1.8454	1.1235	-1.6425	2	0.2422	Egger's test non-significant (p = 0.242); k = 4 provides insufficient power to detect asymmetry
FBG	NR	2	Not generated (k < 3)						k = 2: insufficient for publication bias assessment
FFM_pct	NR	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
HDL	NMN	3	funnel_HDL_NMN_vs_PBO.png	-1.0714	2.8920	-0.3705	1	0.7741	Egger's test non-significant (p = 0.774); k = 3 provides insufficient power to detect asymmetry
HDL	NR	3	funnel_HDL_NR_vs_PBO.png	0.3359	1.4442	0.2325	1	0.8545	Egger's test non-significant (p = 0.855); k = 3 provides insufficient power to detect asymmetry

									asymmetry
HOMA-IR	NMN	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
HOMA-IR	NR	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
HbA1c	NMN	2	Not generated (k < 3)						k = 2: insufficient for publication bias assessment
HbA1c	NR	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
IL-6	NR	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
LDL	NMN	3	funnel_LDL_NMN_vs_PBO.png	-0.3060	1.3045	-0.2346	1	0.8533	Egger's test non-significant (p = 0.853); k = 3 provides insufficient power to detect asymmetry
LDL	NR	3	funnel_LDL_NR_vs_PBO.png	2.0928	0.1634	12.8045	1	0.0496	Egger's test significant (p = 0.050), but k = 3 is far below recommended minimum (k >= 10); result unreliable
NAD+	NMN	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
NAD+	NR	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
SBP	NMN	4	funnel_SBP_NMN_vs_PBO.png	-1.6594	0.8928	-1.8587	2	0.2042	Egger's test non-significant (p = 0.204); k = 4 provides insufficient power to detect asymmetry
SBP	NR	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
TC	NMN	2	Not generated (k < 3)						k = 2: insufficient for publication bias assessment
TC	NR	3	funnel_TC_NR_vs_PBO.png	1.7297	1.4257	1.2132	1	0.4389	Egger's test non-significant (p = 0.439); k = 3 provides insufficient power to detect asymmetry
TG	NMN	4	funnel_TG_NMN_vs_PBO.png	-0.7093	0.4083	-1.7370	2	0.2245	Egger's test non-significant (p = 0.225); k = 4 provides insufficient power to detect asymmetry
TG	NR	3	funnel_TG_NR_vs_PBO.png	0.3611	2.7713	0.1303	1	0.9175	Egger's test non-significant (p = 0.918); k = 3

									provides insufficient power to detect asymmetry
body_fat_pct	NR	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
body_weight	NMN	3	funnel_body_weight_NMN_vs_PBO.png	2.1506	0.8237	2.6108	1	0.2329	Egger's test non-significant (p = 0.233); k = 3 provides insufficient power to detect asymmetry
body_weight	NR	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
fasting_insulin	NMN	2	Not generated (k < 3)						k = 2: insufficient for publication bias assessment
fasting_insulin	NR	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
fat_mass	NMN	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
skeletal_muscle	NMN	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment
sleeping_metabolic_rate	NR	1	Not generated (k < 3)						k = 1: insufficient for publication bias assessment

Note: Egger's regression test for funnel plot asymmetry. Tests with k < 3 have insufficient power and are not interpretable. Funnel plots generated for all comparisons with k >= 3.